

EASY (10)

1.

What is Hyper-V?

- A) Microsoft's antivirus software
- B) Microsoft's hypervisor for virtualization ☒
- C) A storage management tool
- D) A file system

Answer: B

Explanation: Hyper-V is Microsoft's hypervisor used to create and manage virtual machines.

2.

Which PowerShell command installs Hyper-V role?

- A) Add-WindowsFeature HyperV
- B) Install-WindowsFeature Hyper-V ☒
- C) Enable-HyperV
- D) New-HyperVRole

Answer: B

Explanation: Install-WindowsFeature -Name Hyper-V installs Hyper-V on Windows Server.

3.

What type of disk uses traditional partition tables?

- A) Basic Disk ☒
- B) Dynamic Disk

- C) Storage Pool
- D) Virtual Disk

Answer: A

Explanation: Basic disks use MBR or GPT partition tables.

4.

Which volume type provides data mirroring?

- A) Spanned Volume
- B) Striped Volume
- C) Mirrored Volume ☒
- D) Simple Volume

Answer: C

Explanation: Mirrored volume duplicates data on two disks (RAID 1).

5.

What is the purpose of Storage Spaces?

- A) To create virtual disks from physical disks ☒
- B) To partition disks
- C) To format disks
- D) To create snapshots

Answer: A

Explanation: Storage Spaces pools physical disks into virtual disks.

6.

What file system is designed for high resilience and scalability?

- A) FAT32
- B) NTFS
- C) ReFS ☒
- D) exFAT

Answer: C

Explanation: ReFS is designed for reliability and scalability in server storage.

7.

What does data deduplication do?

- A) Creates backups
- B) Deletes duplicate files
- C) Eliminates duplicate data chunks to save space ☒
- D) Compresses files

Answer: C

Explanation: Deduplication saves space by storing only unique data chunks.

8.

Which volume type stripes data without redundancy?

- A) Mirrored Volume
- B) Spanned Volume
- C) Striped Volume ☒
- D) Simple Volume

Answer: C

Explanation: Striped volume (RAID 0) spreads data across disks for performance without fault tolerance.

9.

Which Hyper-V feature allows saving a VM's state?

- A) Snapshots (Checkpoints) ☒
- B) Storage Pools
- C) Dynamic Disks
- D) Data Deduplication

Answer: A

Explanation: Checkpoints save the current state of a VM.

10.

Which disk type supports volumes spanning multiple disks?

- A) Basic Disk
- B) Dynamic Disk ☒
- C) SSD
- D) Virtual Disk

Answer: B

Explanation: Dynamic disks support spanned and mirrored volumes.

MEDIUM (20)

11.

What type of virtual disk does Hyper-V support natively?

- A) VHD and VHDX
- B) ISO
- C) VMDK
- D) IMG

Answer: A

Explanation: Hyper-V uses VHD and VHDX virtual hard disks.

12.

Which tool is NOT used to manage disks and volumes?

- A) Disk Management
- B) PowerShell
- C) Registry Editor
- D) Storage Spaces

Answer: C

Explanation: Registry Editor is not used for disk management.

13.

Which PowerShell cmdlet creates a new partition?

- A) New-Disk

- B) New-Partition ☒
- C) New-Volume
- D) Initialize-Disk

Answer: B

Explanation: New-Partition creates a new partition on a disk.

14.

In Storage Spaces, what resiliency option protects against a single drive failure?

- A) Simple
- B) Mirror ☒
- C) Parity
- D) None

Answer: B

Explanation: Mirror provides redundancy by copying data.

15.

Which Hyper-V feature allows virtual machines to communicate over a network?

- A) Virtual Switch ☒
- B) Storage Pool
- C) Checkpoints
- D) Data Deduplication

Answer: A

Explanation: Virtual Switch connects VMs to networks.

16.

What is a limitation of data deduplication?

- A) Cannot be used on ReFS volumes
- B) Only works on NTFS or ReFS volumes ☒
- C) Cannot deduplicate files larger than 100MB
- D) Only works on SSDs

Answer: B

Explanation: Deduplication works only on NTFS or ReFS.

17.

Which volume type uses striping with parity?

- A) Mirrored Volume
- B) RAID 5 Volume ☒
- C) Simple Volume
- D) Spanned Volume

Answer: B

Explanation: RAID 5 uses striping with parity for redundancy.

18.

What happens when you enable data deduplication?

- A) Files are compressed
- B) Duplicate data chunks are removed and replaced with pointers ☒

- C) Files are deleted
- D) Volumes are formatted

Answer: B

Explanation: Deduplication replaces duplicate chunks with references.

19.

Which Hyper-V generation supports UEFI boot?

- A) Generation 1
- B) Generation 2 ☒
- C) Both
- D) Neither

Answer: B

Explanation: Generation 2 VMs support UEFI boot.

20.

Which of the following is NOT a type of Storage Spaces resiliency?

- A) Simple
- B) Mirror
- C) Parity
- D) RAID 10 ☒

Answer: D

Explanation: RAID 10 is not a Storage Spaces resiliency option.

21.

Which PowerShell command enables data deduplication on volume D:?

- A) Enable-DedupVolume -Volume D: ☒
- B) Start-Dedup -Volume D:
- C) New-Dedup -Volume D:
- D) Set-Dedup -Volume D:

Answer: A

Explanation: Enable-DedupVolume enables deduplication.

22.

Which file system is NOT supported by Windows Deduplication?

- A) NTFS
- B) ReFS
- C) FAT32 ☒
- D) All are supported

Answer: C

Explanation: Deduplication does not support FAT32 volumes.

23.

What is a spanned volume?

- A) A volume that copies data across disks
- B) A volume that combines free space of multiple disks into one large volume ☒
- C) A volume striped for performance
- D) A volume with parity

Answer: B

Explanation: Spanned volumes combine disks into a large logical volume.

24.

What PowerShell cmdlet lists all disks?

- A) Get-Volume
- B) Get-Disk ☒
- C) Get-Partition
- D) List-Disks

Answer: B

Explanation: Get-Disk lists physical disks.

25.

Which Hyper-V feature allows live migration of VMs?

- A) Checkpoints
- B) Virtual Switch
- C) Failover Clustering ☒
- D) Data Deduplication

Answer: C

Explanation: Failover Clustering supports live migration.

26.

What is the default maximum size of a VHDX file?

- A) 2 TB
- B) 16 TB ☒
- C) 1 TB
- D) 512 GB

Answer: B

Explanation: VHDX supports up to 16 TB size.

27.

Which file system is best suited for large volumes and resiliency in Windows Server?

- A) FAT32
- B) NTFS
- C) ReFS ☒
- D) exFAT

Answer: C

Explanation: ReFS is designed for large resilient volumes.

28.

What is the first step to enable data deduplication on a server?

- A) Format the volume
- B) Enable the Data Deduplication role or feature ☒
- C) Create a storage pool
- D) Install Hyper-V

Answer: B

Explanation: You must enable the Data Deduplication feature in Server Manager or PowerShell.

29.

Which of the following is NOT a valid disk initialization style?

- A) MBR
- B) GPT
- C) FAT32 ☐
- D) None of these

Answer: C

Explanation: FAT32 is a file system, not a disk initialization style.

30.

What is required to create a mirrored volume on Windows?

- A) Two basic disks
- B) Two dynamic disks ☐
- C) Two storage pools
- D) Two USB drives

Answer: B

Explanation: Mirrored volumes require dynamic disks.

HARD (10)

31.

When using Storage Spaces, what resiliency level requires at least three disks?

- A) Simple
- B) Mirror
- C) Parity ☒
- D) None

Answer: C

Explanation: Parity resiliency requires at least three disks for parity calculations.

32.

Which Windows Server role is mandatory for WDS but optional for Hyper-V?

- A) DHCP
- B) File Server
- C) Hyper-V
- D) None (Hyper-V role is independent) ☒

Answer: D

Explanation: Hyper-V is an optional role unrelated to WDS.

33.

Which command shows the deduplication status of a volume?

- A) Get-DedupStatus -Volume D: ☒

- B) Check-Dedup -Volume D:
- C) Show-Dedup -Volume D:
- D) Status-Dedup -Volume D:

Answer: A

Explanation: Get-DedupStatus shows deduplication status.

34.

How does Hyper-V implement virtual networking isolation?

- A) VLAN tagging on virtual switches ☒
- B) Separate physical NICs only
- C) Using checkpoints
- D) Data deduplication

Answer: A

Explanation: Hyper-V uses VLAN tagging to isolate VM networks.

35.

What is the main limitation of mirrored volumes on dynamic disks?

- A) No fault tolerance
- B) Requires double disk space for redundancy ☒
- C) Cannot be resized
- D) Only supports FAT32

Answer: B

Explanation: Mirrored volumes store a full copy on both disks.

36.

Data deduplication does NOT support which file attribute?

- A) Compressed files
- B) Encrypted files ☒
- C) Sparse files
- D) Read-only files

Answer: B

Explanation: Deduplication is not supported on encrypted files.

37.

What is the default file age threshold before deduplication processes a file?

- A) 0 days
- B) 3 days ☒
- C) 7 days
- D) 30 days

Answer: B

Explanation: By default, files must be older than 3 days.

38.

In Hyper-V, what is a Generation 2 VM's default secure boot template?

- A) Microsoft Windows ☒
- B) Microsoft UEFI

- C) Microsoft Legacy BIOS
- D) None

Answer: A

Explanation: Generation 2 VMs have secure boot enabled with Microsoft Windows template.

39.

Which feature allows data deduplication to run automatically on a schedule?

- A) Deduplication Scheduler ☒
- B) Volume Shadow Copy
- C) Checkpoints
- D) Storage Pools

Answer: A

Explanation: Deduplication runs via scheduled jobs.

40.

Which is true about Storage Spaces Direct?

- A) Requires at least 2 nodes ☒
- B) Runs only on single server
- C) Only supports SATA disks
- D) Does not support deduplication

Answer: A

Explanation: Storage Spaces Direct is a clustered storage solution requiring multiple nodes.